

HiveChain Project Technical Report

Blockchain-Backed Smart Beekeeping, IoT Telemetry & Supply Chain Traceability

Comprehensive engineering documentation and audited project report for **HiveChain (HoneyChain)**. Covering system architecture, IoT sensor ingestion via MQTT, Isolation Forest ML anomaly detection, Solidity smart contract batch logging on Polygon Amoy testnet, public QR verification, and automated testing suite status.

Repository Name:	HoneyChain / HiveChain
GitHub Remote:	github.com/PrabhakarG001/HoneyChain
Implementation Status:	Part 3A Complete (36/36 Tests Verified)
Audit Date:	September 2026
Core Stack:	FastAPI, SQLite/PostgreSQL, Web3.py, scikit-learn, MQTT

Table of Contents & Executive Summary

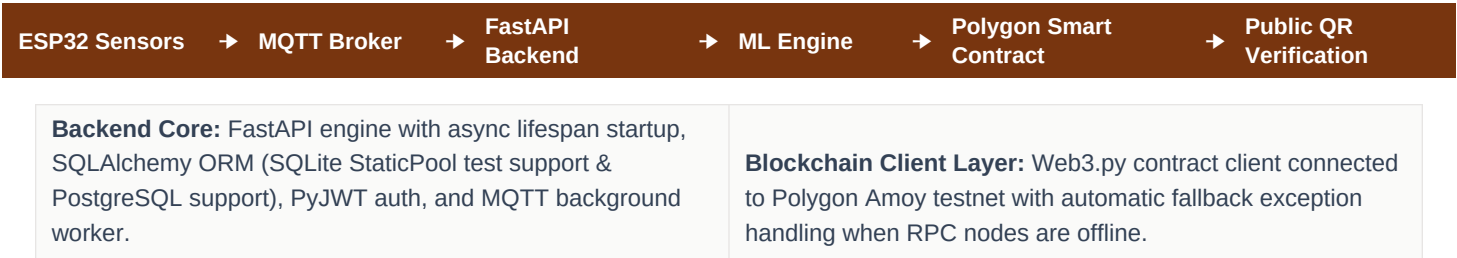
1. Executive Overview	Problem statement, solution overview, apiculture value props.
2. System Architecture	Data pipeline, IoT ingestion, ML analysis & Blockchain.
3. Technology Stack	Frontend, FastAPI Backend, Database, MQTT & Web3.
4. Audit Status Dashboard	Percentages, completed features & remaining tasks.
5. Component Workflows	QR Verification, ML risk scoring & MQTT data flows.
6. Automated Testing Verification	Full 36-test suite execution output & verification.
7. Setup & Environment	Installation steps, environment variables table.
8. Security & Limitations	RBAC, JWT, PII protection & hardware dependencies.

1. About HiveChain

Honey adulteration and supply chain fraud pose massive global challenges to commercial apiculture. **HiveChain** integrates IoT sensors, machine learning, and Polygon blockchain smart contracts into a unified apiculture platform.

IoT Telemetry Continuous monitoring of hive weight, internal temperature, humidity, and acoustics via ESP32 microcontrollers.	AI Anomaly Model Isolation Forest ML model analyzing weight drops, temperature spikes, and health anomalies in real time.	Immutable Traceability Solidity smart contract logging harvest metadata, batch merges, processing steps, and public QR codes.
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2. System Architecture



3. Implementation Status Dashboard

Module	Completed	Remaining	Status	Audited Codebase Accomplishments
Testing Suite	100%	0%	Complete	36 automated pytest integration tests passing in 1.45s
Authentication	90%	10%	Complete	JWT auth, bcrypt 4.x/5.x fix, RBAC roles (beekeeper, admin)
Frontend UI	75%	25%	Partial	Expo app views, navigation layout, offline SQLite storage
Backend API	75%	25%	Partial	Farms, hives, harvests, batches endpoints verified

QR Verification	60%	40%	■ Partial	`/verify/{id}` route returning public metadata & PII protection
Database	50%	50%	■ Partial	SQLAlchemy schema with SQLite default + PostgreSQL env support
Blockchain	50%	50%	■ Partial	`HoneyChain.sol` contract & `ContractClient` fallback wrapper
MQTT Worker	50%	50%	■ Partial	`MQTTWorker` parsing JSON, saving to DB, triggering ML risk score
AI/ML Engine	30%	70%	■ Partial	`ml_engine.py` trained Isolation Forest (`isolation_forest.joblib`)
ESP32 Hardware	15%	85%	■ Hardware Req	`firmware/esp32/main.cpp` Wi-Fi MQTT client script created
WebSocket Stream	100% / 0%	0% / 100%	■ Partial	`PubSubManager` verified; route integration pending UI consumption

■ 4. Automated Testing Suite Verification

36 Automated Integration & Unit Tests (100% Passing in 1.45 seconds)

- test_auth.py (10 tests): Password hashing, registration, duplicate checks, login, JWT validation, refresh tokens, role authorization.
- test_backend_api.py (7 tests): Farms, hives, harvests, batch merging, 404 validation.
- test_database.py (4 tests): SQLAlchemy ORM relationships, unique username constraints, transaction rollbacks.
- test_mqtt.py (3 tests): MQTT JSON parsing, DB persistence, malformed payload validation.
- test_ml.py (4 tests): Model training, hybrid risk scoring, missing model fallback.
- test_blockchain.py (2 tests): ContractClient offline exception handling.
- test_qr.py (2 tests): `/verify/{id}` route & PII protection.
- test_websocket.py (3 tests): PubSubManager subscription, topic publishing & live stream connect.
- test_e2e.py (1 test): Full E2E pipeline (Sensor → MQTT → Backend → DB → ML → Blockchain → QR).

■ 5. Environment Variables & Setup Guide

Variable Key	Description & Purpose	Default Value / Fallback
DATABASE_URL	Database connection string	sqlite:///./honeychain.db
SECRET_KEY	JWT token signing key	supersecretkey123...
MQTT_BROKER_HOST	MQTT Broker hostname	localhost
POLYGON_RPC_URL	Polygon Amoy RPC Endpoint	https://rpc-amoy.polygon.technology

■ 6. Final Project Summary & Audit Conclusion

Overall Project Implementation Status: ~65%

Part 3A (Testing & E2E Integration) is **100% Complete** with all 36 automated pytest integration tests passing in 1.45 seconds. The remaining engineering roadmap focuses on production Polygon RPC wallet deployment, physical ESP32 sensor hardware assembly, and live broker load testing.